HARDWARE APPLICATION

MAKING THE SHIFT KEY MODIFICATION

When the Apple was first put together, its designer could hardly have envisioned the variety of purposes for which his creation would be used. He obviously did not envision word processing as an Apple task, or he would have included an upper/lower case keyboard and an 80 column text display. Undaunted, thousands of people use the Apple for word processing anyway, and make the Apple an acceptable word processing computer by installing an 80 column video board and the SHIFT key modification.

The SHIFT key mod was the bright idea of some forgotten soul who noted that the PUSHBUTTON 2 input to the game I/O socket was rarely used. He connected the SHIFT key output to pin 4 of the game I/O socket and interpreted PUSHBUTTON 2 as determining upper or lower case for alphabetic input from the keyboard. This modification has become significant because most word processing programs support it, allowing persons to enter upper and lower case to word processors via the SHIFT keys. The monitor GETLN routine, of course, does not support the SHIFT key mod, so BASIC still will communicate with the keyboard only in upper case.

The SHIFT key mod will not work if a pulled down pushbutton is also connected to the PUSHBUTTON 2 input. This is because the SHIFT key line is pulled up, and you cannot wire-OR a pulled up line and a pulled down line. For this reason, some joysticks with three buttons will be incompatible with the SHIFT key mod.

The SHIFT key mod is most easily installed on the Apple II Plus. This is because the SHIFT' line is highly accessible on the exposed pins of PLUG 1, which connects the main keyboard PC card to the keyboard encoder card . This plug can be seen by removing the Apple's cover and looking in from the rear. The SHIFT' line is pin 24, the second pin from the left. Take a wire which is terminated on both ends with a spring loaded clip. With the computer turned off, clip one end to pin 24 of PLUG 1. Remove your joystick or paddle set from the game I/O socket and clip the other end of the jumper to pin 4 on the joystick plug. Reinsert the joystick or paddle set with the clip attached and you're finished.

Here is a method which takes a little more work but is more usable. Take a wire which is terminated on one end with a spring loaded clip. Solder the other end to the base of pin 4 on a high quality 16-pin DIP socket. Clip the spring clip to pin 24 of PLUG 1 and insert the socket into the game I/O connector. Other devices may then be inserted into the socket.

The SHIFT key mod is more difficult to install on the older keyboards predating the Apple II Plus. There is no convenient exposed point on the SHIFT' line to which a spring loaded clip can be attached. You may solder a wire directly to Key 42 or Key 53 and connect the other end to pin 4 of the game I/O socket. Whatever connection method is used should allow the jumper to be easily disconnected in case the white enclosure needs to be separated from the base assembly. The method shown here is to connect the SHIFT' line to the game I/O socket using a spare line on the 16-pin keyboard connector. This takes a little effort, but when you're finished the SHIFT key mod is fully integrated with the Apple mechanical package with no loose wires to be connected or disconnected. Here is the installation procedure:*

1. Separate the bottom assembly from the white case following the procedure in Appendix J.
2. Remove the keyboard from the white case by removing the three screws on either end.
3. Lay the keyboard face down and solder an insulated wire jumper between Key 53 and pin 4 of the keyboard connector as shown in Figure 7.18.
4. Remove the motherboard from the base assembly following the procedure in Appendix J.
5. Place the motherboard bottom side up and solder an insulated wire jumper between pin 4 of the game I/O socket and pin 4 of the keyboard socket as shown in Figure 7.18. This wire can be held tightly to the motherboard with spots of glue. A hot glue gun works well for this purpose.
6. Reinstall the motherboard on the base assembly.
7. The installation is complete. Operation can be verified by setting the keyboard on an insulated surface next to the base assembly, connecting the keyboard jumper to the motherboard, and applying power. After verification, disconnect the keyboard from the motherboard, install the keyboard in the white case, and install the base assembly in the white case. Don't forget to reconnect all cards and wires, including the keyboard jumper.

*Please read the NOTE OF CAUTION at the beginning of the book before making any modification to your hardware.

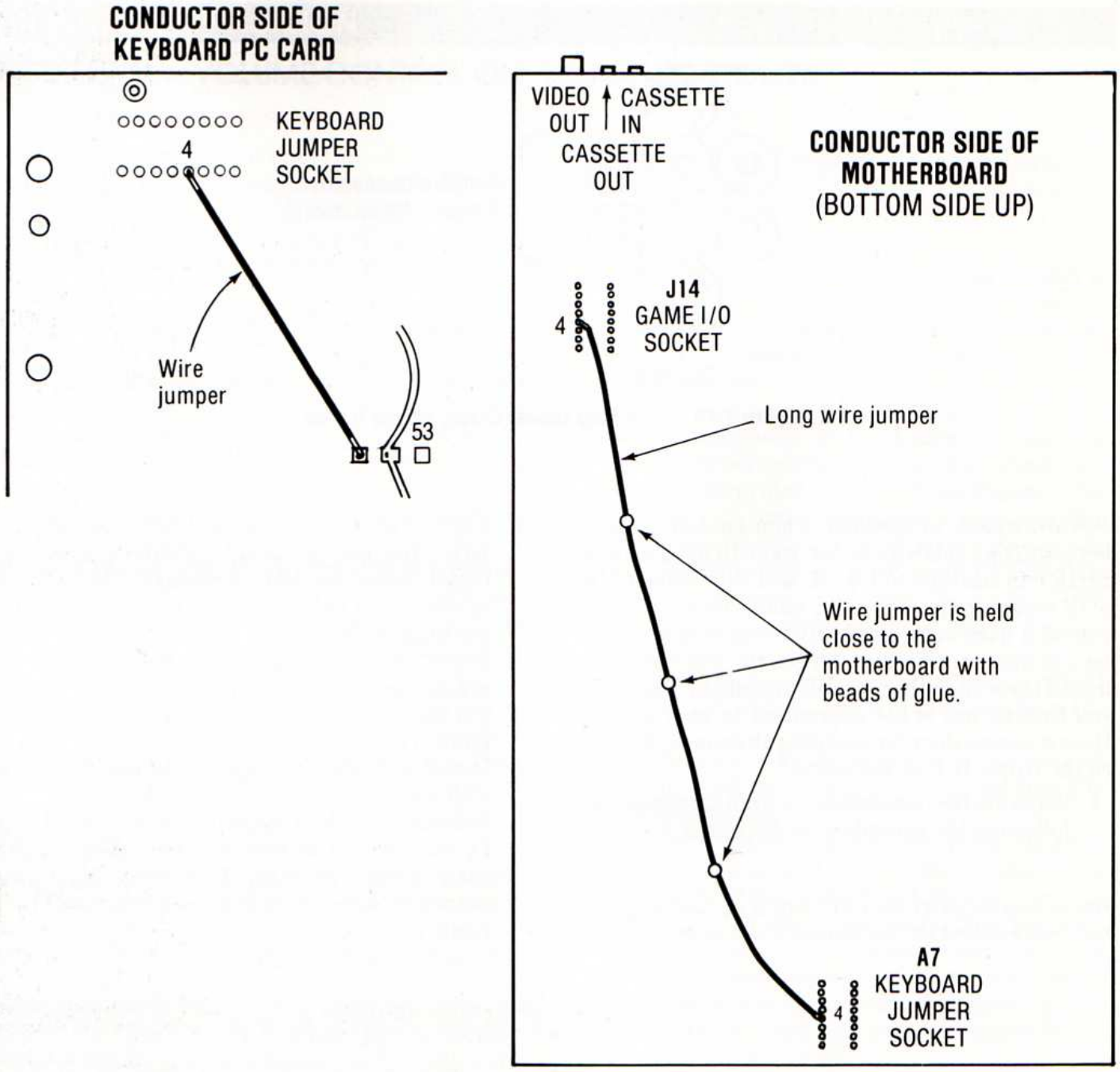


Figure 7.18 Installation of SHIFT Key Mod, for Older Keyboards.

There is an alternate method to the SHIFT key mod for obtaining the upper/lower case alphabet from the keyboard in the Apple II Plus. The keyboard encoder ROM is fully programmed for upper and lower case in the II Plus, and there are holes and connections for a slide switch to enable or disable alphabetic shifting. The switch connections are normally jumpered so alphabetic shifting is disabled. You can change the configuration so that alphabetic shifting is enabled, or you can mount a slide switch and disable or enable alphabetic shifting at your pleasure. Even though the keyboard will output upper and lower case ASCII to the MPU, the monitor GETLN routine assumes everything is upper case only. The keyboard will communicate in upper and lower case only to the limited number of programs written for the Apple which look for lower case alphabetic ASCII in the keyboard input. Your word processor may or may not support this capability in addition to the more prevalent SHIFT key mod.